Diffusion of Innovation and Fitting Sigmoid Curves

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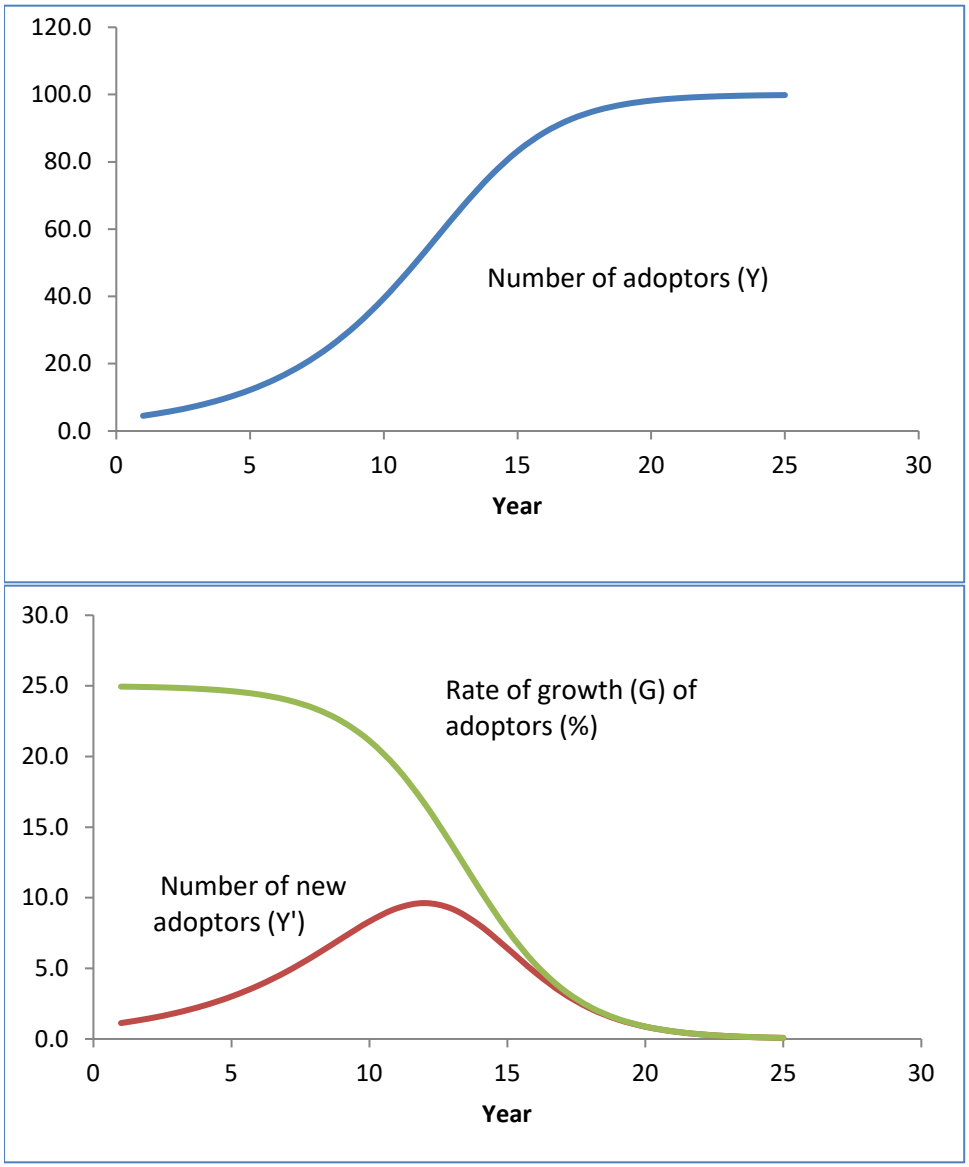
This technical note covers the use of sigmoid or "S-shaped" growth models in forecasting. Such models apply when there are limits to growth so that initial high rates of growth slow as the carrying capacity of the economic or biological environment is reached.

Exhibit 1—Sigmoid Curves



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A quantity Y is observed over time with $Y(x)$ being its value at time *x*. Denote its rate of increase at time *x* as $Y'(x)$. Formally, $Y'(x)$ is the derivative of $Y(x)$ or the slope of the curve at time *x*. In the literature on sigmoid growth this rate of expansion Y' is often called the "rate of growth." It is not. The growth rate of Y is the ratio of Y' to Y. Here the growth rate of Y is denoted as $G_Y = Y'/Y$. If the rate of growth is constant we say that the growth is exponential. In this case $Y' = rY$ so that the rate of expansion is proportional to the current level *Y*. The solution to this differential equation is the standard expression for exponential growth:

$$Y(x) = Y_0 e^{rt}. \tag{1}$$

Sigmoid growth occurs at rate which declines over time. Different curves arise out of different assumptions about how and why this happens. The curve presented here is called the Richards curve or, sometimes, the generalized logistic. The commonly used logistic and Gompertz curves are special cases of the Richards curve.[1]

The Richards law of expansion is

$$Y' = \frac{r}{b} Y \left[1 - \left(\frac{Y}{K} \right)^b \right]. \tag{2}$$

The parameter *K* is the ultimate level of *Y*—the carrying capacity of the system. To understand this formula, note that if $Y = 0$ then $Y' = (r/b)Y$, a pattern of exponential growth rate r/b. On the other hand, as Y rises closer to *K*, then $Y' \to 0$, putting a stop to further expansion. The parameter *b* determines the skewness or asymmetry of the curve. The parameter *b* must be larger than zero.[2] The curve shown in Exhibit 1 is a Richards curve with $b = 2$.

Solving the Richards differential equation gives the Richards curve:

$$Y(x) = \frac{K}{\left(1 + b e^{-r(x-M)}\right)^{\frac{1}{b}}} \tag{3}$$

Fitting a Richards Curve to Data

Exhibit 2 displays a time series of the percentage of Japanese steel making capacity that used the oxygen process. This then-new technology began to be installed in 1958 and by 1973 accounted for slightly over 80% of capacity. (The data are available on JapanOxy.xlsx.)

This kind of process is called technological substitution—a new technology (oxygen) is replacing an older technology.

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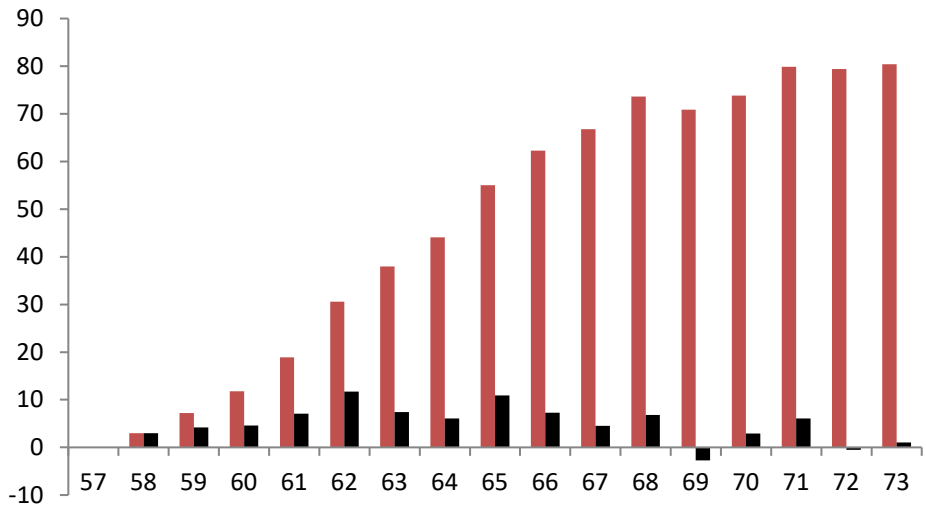
[1] When $b = 1$ the Richards curve reduces to the logistic curve. When *b* is very small, it approaches the Gompertz.

[2] The Richards curve is defined for $-1 < b < 0$, but interpretation is difficult and numerical issues can arise.

Exhibit 2—Percent of Japan Steel Capacity Using Oxygen Process and Annual Changes in the Percent



In general, fitting a sigmoid curve to data is extremely chancy if the inflection point has not yet occurred. Recall that the inflection is when Y', the rate of change in *Y,* peaks. Exhibit 2 also shows the annual changes in percent oxygen capacity—the variable denoted Y' in this note. As can be seen, Y' looks to have peaked in about 1962, marking the probable inflection point.

Various statistical programs like SAS, Stata, R, and Matlab have built-in procedures for fitting non-linear models like the Richards curve. Here I shall work with Excel for two reasons. First, everyone has access to it. Secondly, the best way to learn about curve-fitting is to do it "by-hand." The easiest way to lean this is to follow along using Excel.

To begin, the original data consists of two columns: the first contains years and the second the percentage of oxygen capacity. It is good practice to label each column with a unique name and use *Formulas/CreateFromSelection* to assign these names to the columns of data below. Here they are labeled Year and Yobs. Add a third column labeled Yprime (for Y') and fill it with the year-to-year differences in Yobs (the will normally be unavailable but here it is zero.)

Above the data, create parameter block with two rows. Put the parameter names in the first row and initial estimates in the second. Use *Formulas/CreateFromSelection* to name each parameter cell. The four parameters of the Richards curve are *K*, *r*, *M*, and *b*. Excel doesn't like single-character parameter names or names that look like cell references. Hence, double character names are used here: KK, rr, MM, and rr. The whole row of four parameters is labeled param**s**. Reasonable initial values for rr and rr are 0.2 and 1.0. Set the initial guess at the inflection point MM at 1962. Set the initial value of the saturation level KK at 85. .

Using these parameter values, create a new column of Richards curve predictions labeled Yfit. Compute Richards curve values for each year either by using formula (3) directly or with the RICHARDS function provided with ASA.XLAM.

Below the parameter block, create a block of summary statistics. Put these statistics over the Yfit column. The first item is SSTm—the sum of squares of the data corrected for the mean. The easy way to get this number is to use Excel formula COUNT(Yobs)*VARP(Yobs). Below that enter SSE—the sum of squared errors. Here use Excel array formula SUM(Yobs-Yfit)^2). Below that

Exhibit 3

	A	B	C	D	E
1	**TechNote 3: Adoption of Oxygen Process in Japan**				
2					
3		KK	rr	bb	MM
4	***Params***	85.0	0.20	1.00	1962.0
5					
6	SSTm		14050.29		
7	SSE		2934.363		
8	R^2		0.791153		
9					
10	***Year***	***Yobs***	***Yfit***		
11	1957.00	0	22.86002		
12	1958.00	3	26.35217		
13	1959.00	7.2	30.11921		

compute the R-squared of the fit as 1 – SSE/SSTm. At this point, the upper portion of the sheet should look like Exhibit 3.

To find a "best fit" use Excel's Solver add-in. (Make sure it is checked off in the add-ins list.) We want Solver to find the parameter values which minimize the sum of squared errors or SSE.

Excel Solver: Invoke Solver, give it the target of the SSE number, specify minimization, and enter PARAMS in the "By Changing Cells" box. Click the "Options" button and then check the "Assume Non-Negative" box. This will make sure Solver doesn't try out negative values of rr and bb. Also check the "Use Automatic Scaling" box. Click OK then the "Solve" button. Solver will report a solution—accept it.

Premium Solver: Invoke Premium Solver from the Add-Ins menu on the ribbon/toolbar. Highlight "Objective" and click "Add" and then select the SSE cell to be minimized. Highlight "Varables/Normal," click "Add" and type "params" in the range box. Click the "Model" tab and click the "Excel Interpreter" dot. Close the model tab and click the "Options" button. Check the boxes next to "Assume Non-Negative" and "Use Automatic Scaling." Click "OK" and then "Solve."

Risk Solver Platform: Click the "Risk Solver Platform" entry on the ribbon/toolbar. Click the "Reset All" option to clear any previous work. Select the SSE cell and then click the "Objective/Min/Normal" menu item. Select the "params" cells and click the "Decisions/Normal" menu item. On the" Solver Options and Model Specs" window, click the "Platform" tab. Set the "Interpreter" to "Automatic." In the "Default Bounds" area, set "Decision Vars Lower" to 0. Click the "Engine" tab at the top. Select the "Standard GRG Nonlinear Engine" and set "Use Automatic Scaling" to True. You can now click the "Optimize/Solve Complete Problem" menu item or click the green arrow on the upper right o the "Solver Options and Model Specs" window.

The solution is a pretty good fit with a 99.7% R-squared (SSE = 41.76573.) The best-fit parameters are:

KK	rr	bb	MM
82.87	0.324	0.117	1962.4

There seems to be an inflection point in 1962, as we guessed. The saturation level is forecast to be 82.9%. And, the fitted rr is much smaller than 1.0, reflecting the asymmetry of the new adoptions curve.

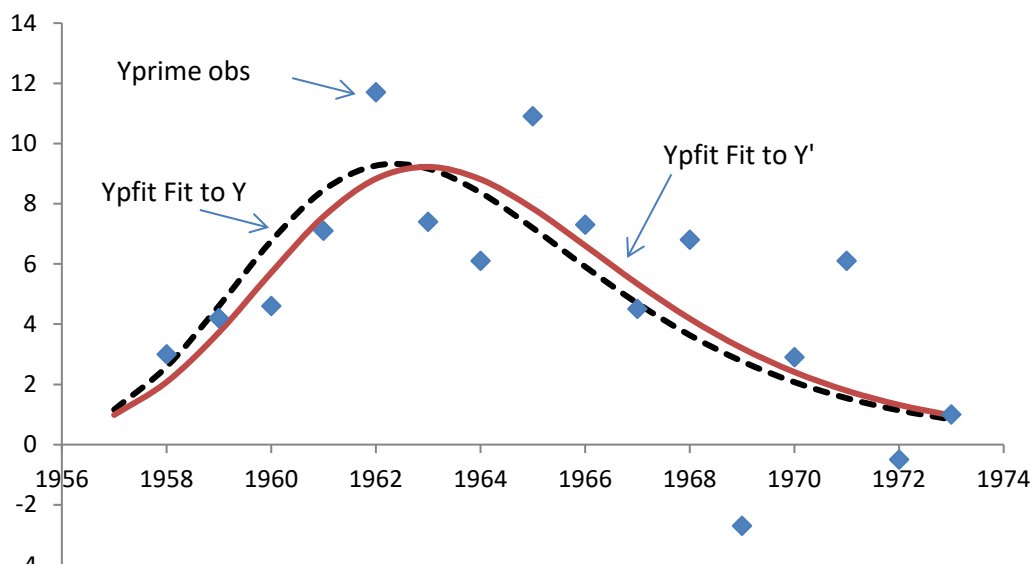
Thinking about the new-adoptions data (Yprime) suggests that we could fit the Richards curve to that instead. To do this, create a new column labeled Ypfit and enter the formula from (2) as (rr/rr)*Yfit*(1-(Yfit/KK)^rr). Or, use the function RICHARDSPRIME in ASA.XLAM. Above the Ypfit column enter a new block of statistics (SSTm, SSE, and R-squared). This time SSTm=COUNT(Yprime)*VARP(Yprime) and SSE = SUM((Yprime-Ypfit)^2). Now, invoke solver and have it minimize this new SSE.

The new solution is slightly different:

	KK	rr	bb	MM
Params	83.75	0.329	0.203	1963.0

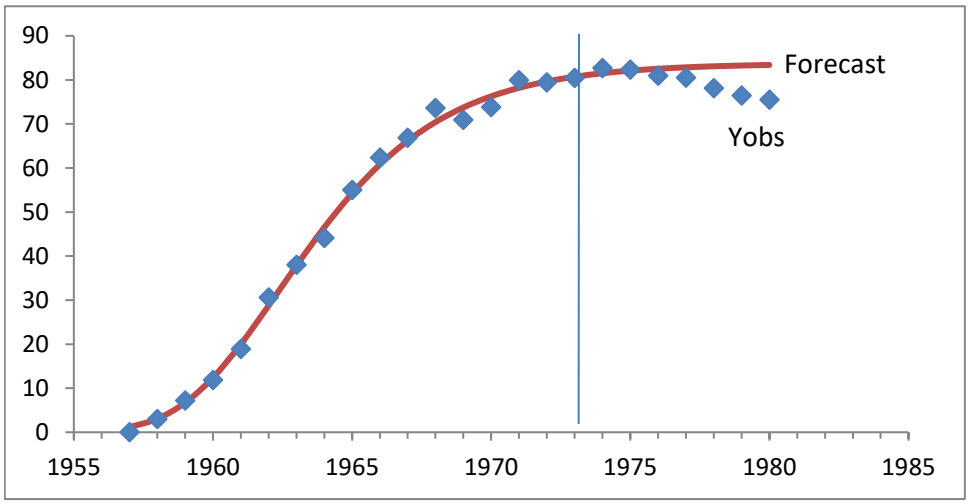
Exhibit 4 shows the difference between the two approaches to fitting the data. They are different because the errors being minimized are different. Do pick an approach it is necessary to examine the errors and decide which one are most significant. Here, fitting Y tends to emphasize later errors because Y is growing over time. On the other hand, fitting to Y' tends to emphasize errors about

Exhibit 4—Fits to Y and Yprime



the time of inflection—in the mid-years of this data. Here I prefer the fit to Y because I am more interested in fitting the later data.

Exhibit 5—Forecast versus Actual



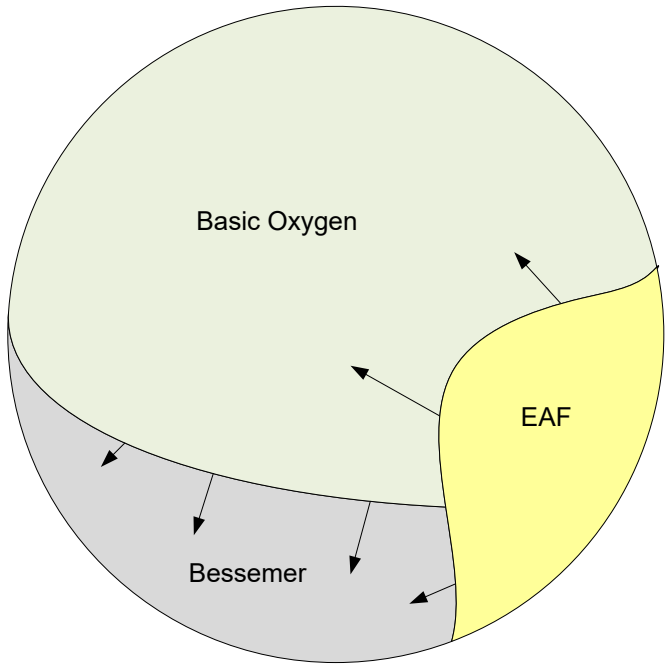
In cases where the value of *Y* increases dramatically it may be useful to compute logs of *Y* and Yfit and minimize the sum of squared errors in logs. This will weight earlier observations more heavily than using raw values. On the other hand, one may prefer to weight more recent values more heavily and stick with the raw values. Curve fitting is an exercise in both technique and judgment.

How well did this fit predict the future? Exhibit 5 shows the fitted Richards curve extended forward into the future and the actual data through 1980. The predicted saturation level of 83% was quite accurate. However, rather than remaining at the saturation level, the percentage of oxygen capacity began to decrease!

This happened because an even newer technology—the electric arc furnace—began to grow in the late 1960s. Thus, while the basic-oxygen process was displacing the older Bessemer (open hearth) process, the electric-arc furnace was beginning a cycle of displacing both.

This dynamic of double substitution is illustrated in Exhibit 6. New technologies and products displace older versions and, in turn, are themselves displaced by newer ones. The complexities arise out of timing and degree. Many factors influence the speed of diffusion of a new technology or product (the r/b ratio). And other factors influence whether or not substitution is partial or complete. In the case at hand, EAF will probably stabilize at a share of about 40% (because it us used primarily for processing scrap.)

Exhibit 6—Illustration of Double Substitution



Regular Solver Difficulties

Regular Excel Solver doesn't always solve. The more variables, the more non-linear the problem, the more likely solver will either pop an error or simply give you the wrong answer. The only way to tell if solver is in trouble is to give it alternative starting values and see if it converges to the same or another answer.

Here are some steps you can take to help solver.

1. Scale your problem in advance. Try to give solver numbers close to 1.0 to work with. For example, it is often useful to convert years (e.g., 1998, 1999) into integers such as 1, 2, …
2. First fit changes and slopes. Here, in the Richards curve problem, it is usually a good idea to first fit Y' because it has more shape to fit. You can then, if you want, use this solution as starting values to fit Y.
3. Fit by steps. We know that if $b = 1$ the Richards curve is the simpler logistic curve. Set $b = 1$ and only fit for KK, RR, and MM. After you have that solution, include BB in the set of cells to change.

Statistics

The purpose of curve fitting is to fit a curve closely to data. If the data is too random, it can hardly be "fit" with a curve. Despite this obvious fact, academics have spent a great deal of time trying to generate statistical measures the goodness of a fit. The R-squared we have computed above is just the beginning of such measures.

Solver does not generate statistical measures, but it is possible to run an Excel macro which will estimate such measures from a Solver solution. The macro SSESTATS included in ASA.XLAM

does just this. The macro works by making incremental changes in each parameter and studying the effects on each observation. The results are the estimated standard error of each parameter estimate.

To invoke the macro, use *developer/macros* and type *ssestats* in the box (the macro will not be on the list below, but the RUN button will darken if Excel finds it).